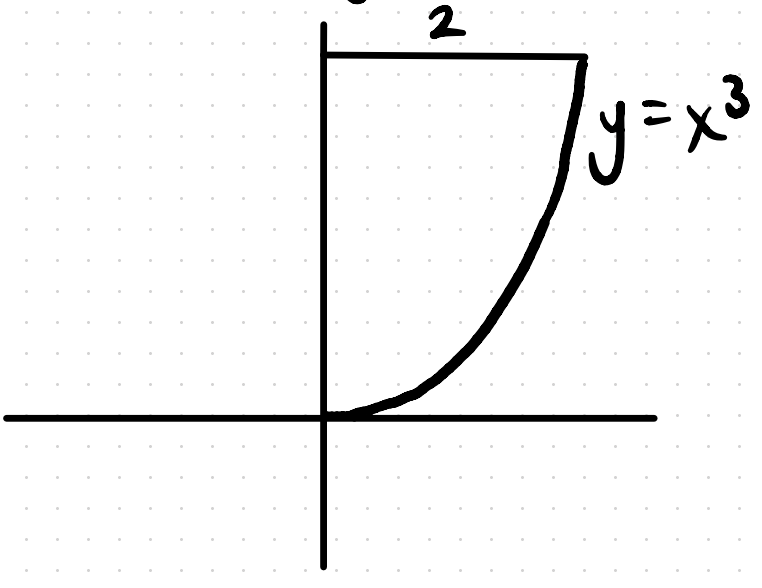
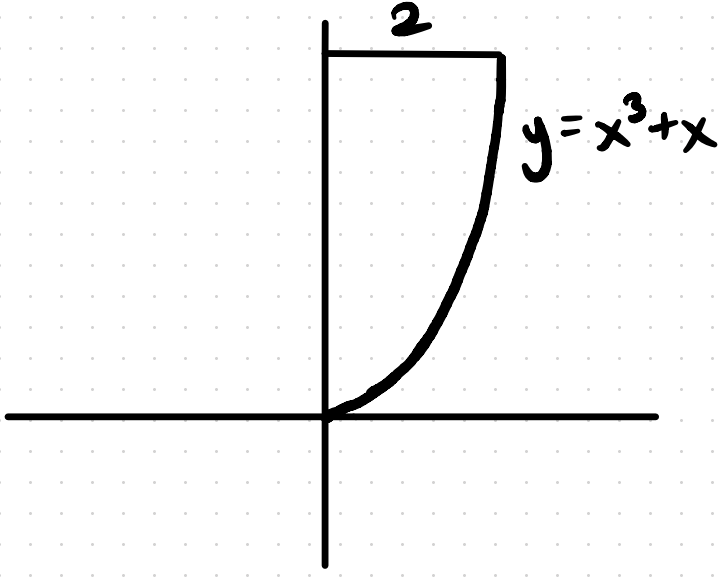
Volumes by cylindrical shells (6.3)

The region bounded by $x=0$, $y=2$, $y=x^3$ is rotated about the y-axis.

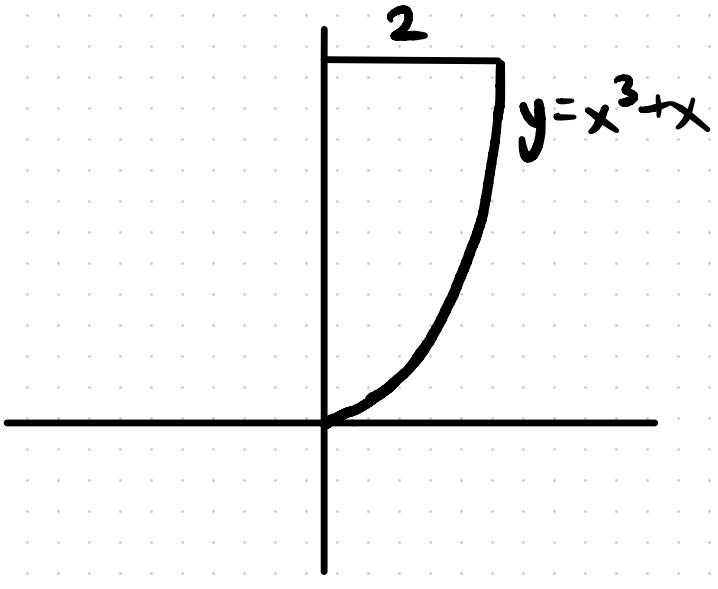


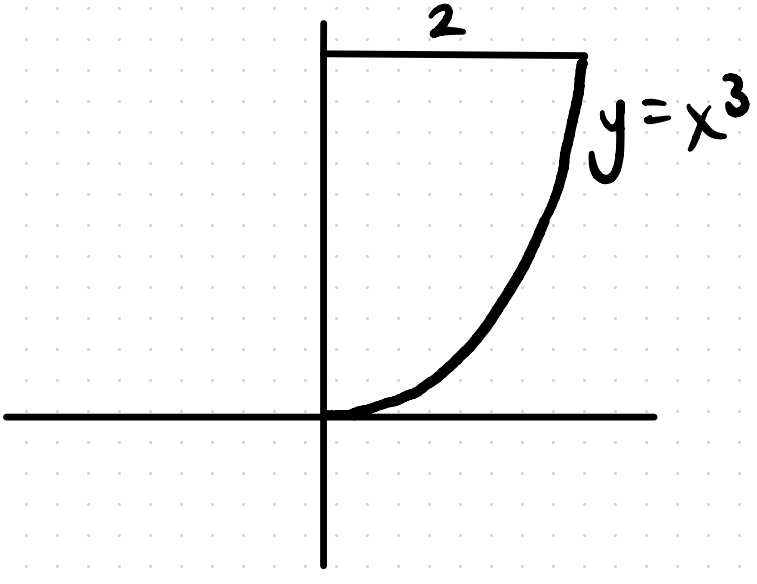
The region bounded by $x=0$, $y=2$, $y=x^3+x$ is rotated about the y-axis.



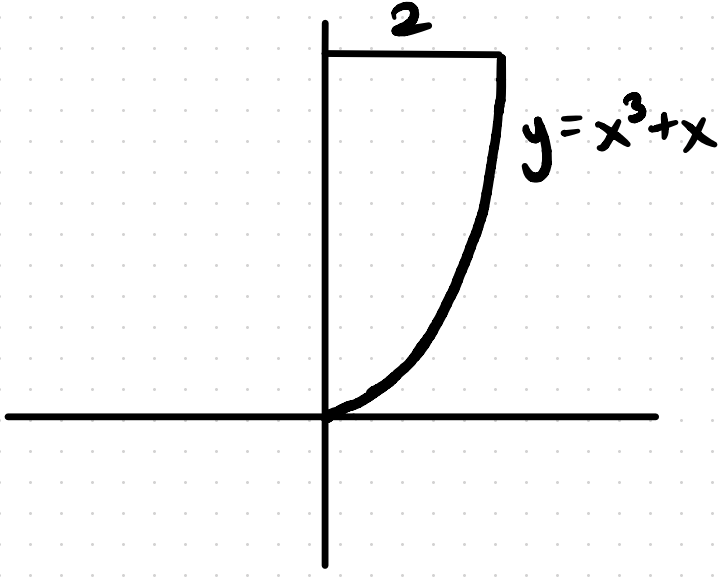
The method of cylindrical shells is a technique used specifically for finding volumes of solids formed by rotating around an axis.

We decompose our solid into hollow cylindrical "shells" with width Δx, where x is the direction perpendicular to the axis of rotation.

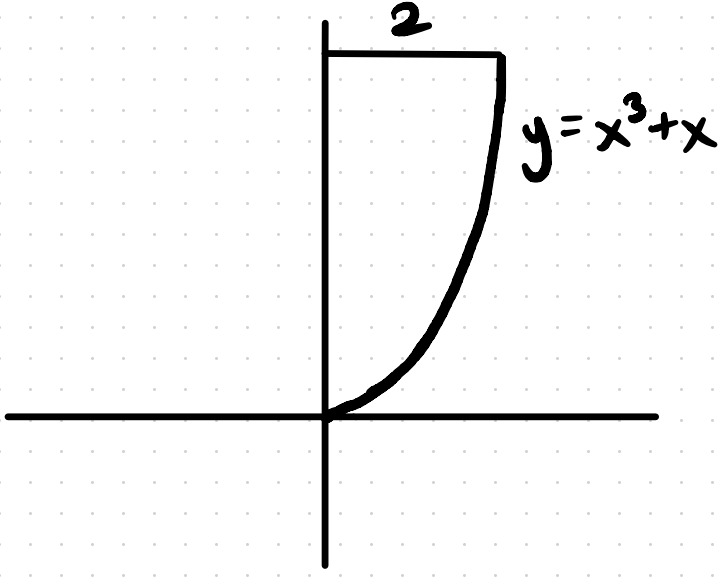

2
$y=x^3+x$


2
$y = x^3$

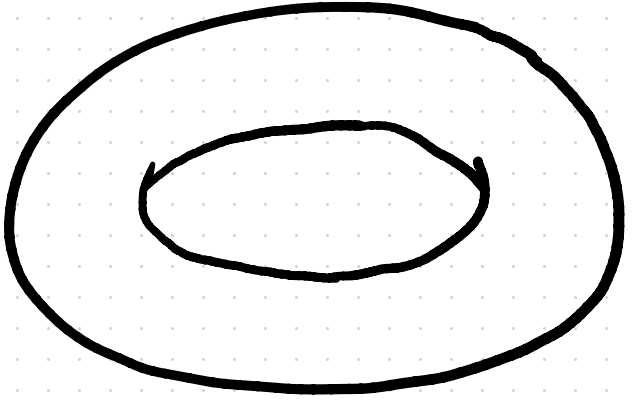
The region bounded by $x=0$, $y=2$, $y=x^3+x$ is rotated about the line $x=-1$.



The region bounded by $x=0$, $y=x+1$, $y=x^3+x$ is rotated about the line $x=-1$



The region bounded by $x = y^5 - y^4$, $x = y^2 - y^3$ is rotated about the line $y = 2$.



Conclusion: Suppose a 2D region R is rotated around a line L, and we want to find the volume of the resulting solid.

